

NoSQL

DATA SCIENCE BOOTCAMP

NASHVILLE  SOFTWARE SCHOOL

Objectives

1. You should be familiar with the terms ACID and BASE as they relate to databases.
2. You should be able to list four types of NoSQL databases.

The ACID Model (Traditional Databases)

ACID ensures **data reliability** in relational databases.

- **Atomic:**
 - A database transaction either succeeds or fails.
 - A transaction cannot be completed only partially.
- **Consistent:**
 - Use of rules and constraints so state is always valid.
 - The data saved can't violate any of the database's integrity.
- **Isolation:**
 - Transactions happen in isolation; No "mid-air collisions."
- **Durability:**
 - Once committed a transaction is permanent, regardless of a subsequent system failure.

Example: During a bank transfer, if one part of the transaction fails, the entire transfer is rolled back.

The BASE Model (NoSQL and Scalable Systems)

BASE prioritizes **scalability and availability** over strict consistency.

- **Basically Available** - The system remains operational and generally available for queries; **no isolation/waiting for other transactions.**
- **Soft State** - Data may change over time
- **Eventually Consistent** - As a system's **state is gradually replicated across all nodes**, it eventually (but not immediately) becomes consistent.

Example: A social media platform displaying the number of likes on a post. Different users may see different values or the value may increase and decrease.

See Tom Scott's video on "Why Computers Can't Count Sometimes":

https://www.youtube.com/watch?v=RY_2qElt3SA

NoSQL – Not only SQL

NoSQL term coined by Carlo Strozzi in 1998 to describe his relational database that did not expose Structured Query Language to users.

What we mean by NoSQL is different (non-relational database), and that term came into use for that meaning around 2009.

Four main types

1. Key-Value Store

- Data is stored as a simple key-value pair (think: Python dictionary)
- No schema; the data can be anything (JSON, text, a binary object)
- Only allows retrieving data by key (fast, but limited)
- Redis, DynamoDB, Riak

2. Document Store

- Stores semi-structured document (eg. JSON, BSON, or XML), where each document has fields and values.
- Flexible schema, where documents in the same collection can have different fields.
- Can query by field values, indexing, and nested data.
- MongoDB, Apache CouchDB, Firebase Firestore

NoSQL – Not only SQL

Four main types

3. Column Store

- Data is stored in column grouped into column families; this improves read performance for analytics.
- Flexible schema; columns can be added dynamically.
- Optimized for fast reads of specific columns.
- Hbase, Cassandra, BigTable

4. Graph

- Best for handling relationships between data points (social networks, recommendation systems, fraud detection, supply chain/logistics)
 - **Nodes:** entities (people, products, locations)
 - **Edges:** relationships between nodes (follows, buys, located in)
 - **Properties:** Metadata attached to nodes/edges (user's name or purchase date)
- Schema-less or flexible schema
- Querying uses graph traversal
- Neo4J, Apache Giraph, AllegroGraph, Virtuoso, InfiniteGraph, FlockDB, etc.

NoSQL – Not only SQL

Data Model ⇄	Performance ⇄	Scalability ⇄	Flexibility ⇄	Complexity ⇄	Functionality ⇄
Key–Value Store	high	high	high	none	variable (none)
Column-Oriented Store	high	high	moderate	low	minimal
Document-Oriented Store	high	variable (high)	high	low	variable (low)
Graph Database	variable	variable	high	high	graph theory
Relational Database	variable	variable	low	moderate	relational algebra

ACID vs. BASE - Key Differences

Feature	ACID (SQL Databases)	BASE (NoSQL Database)
Consistency	Strict	Eventually
Scalability	Harder to scale	Highly scalable
Use Case	Banking, ERP	Social media, real-time analytics
Querying	Relatively easy through JOINS	Often limited

The CAP Theorem (Why NoSQL Exists)

CAP Theorem: A distributed system can have at most two of three:

- **Consistency** – Every node in a distributed cluster returns the same, most recent, and successful write. Every client has the same view of the data.
- **Availability** – Each read and write request for a data item will either be processed successfully or will receive a message that the operation cannot be completed.
- **Partition Tolerance** - System keeps running despite network failures that result in two or more partitions, where the nodes in each partition can only communicate with each other.

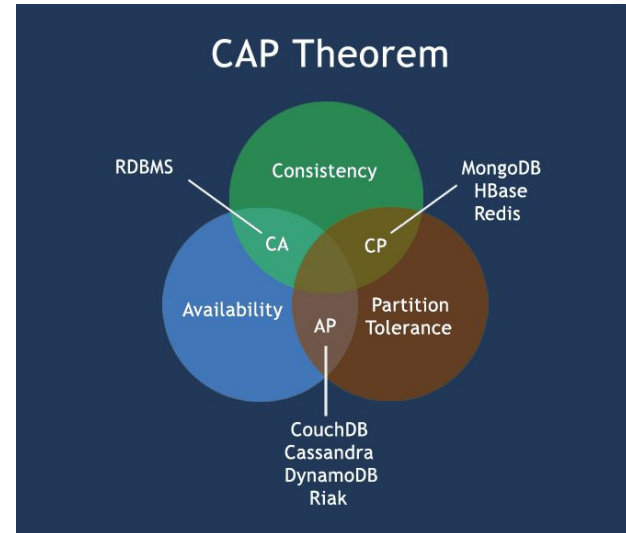


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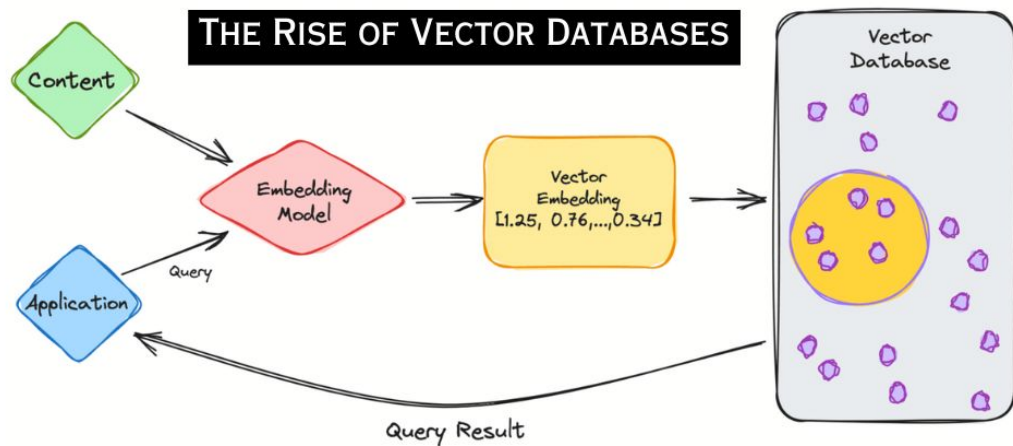
<https://www.geeksforgeeks.org/the-cap-theorem-in-dbms/>

Vector Databases

With the increasing popularity of generative AI, **vector databases** have seen growing use.

Vector databases are optimized to retrieve based on **similarity**.

They are often used as part of a setup called **Retrieval Augmented Generation (RAG)**, as a way to provide additional context to a language model to help it to be able to provide a more relevant answer.



<https://dev.to/pavanbelagatti/wtf-is-a-vector-database-a-beginners-guide-16p>